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A Qualitative Discrepancy between Censored Data Rank Tests

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Summary

Two popular censored data rank tests were used to compare cancer incidence rates between dogs receiving bone marrow transplantation and control dogs. The logrank test gave a significance level of 0.01. In contrast, Gehan's generalized Wilcoxon test gave a significance level of 0.76. The statistics are displayed in a manner that shows how such contrasting significance levels can arise. The Gehan statistic is shown to be subject to a serious criticism that does not apply to other Wilcoxon generalizations. Further insight into the data is obtained using a time-dependent logrank test and the proportional hazards regression model.

1. Introduction

A statistical problem arising in our consulting at the Fred Hutchinson Cancer Research Center (FHCRC) concerned the effect of marrow transplantation and associated total body irradiation (TBI) on cancer incidence in beagle dogs. Basic research in canines plays an important role in undergirding clinical programs in bone marrow transplantation for leukemia and aplastic anemia. All leukemia marrow transplant recipients at the FHCRC receive TBI prior to marrow transplantation. Aplastic anemia patients at the FHCRC no longer receive TBI, although such treatment is still utilized at a number of other institutions.

Cancer incidence and mortality data were collected on 83 beagle dogs receiving TBI and marrow transplantation at the FHCRC, along with similar data on 198 control beagle dogs housed either at the FHCRC or in colony control status at the Argonne National Laboratory. The substantive findings from these data will be discussed elsewhere. The purpose here is to discuss some statistical issues arising in the comparison of data on transplant and control dogs.

Follow-up times varied from less than one year to over 15 years. Cancer incidence times were heavily censored because of non-cancer mortality and more specifically, because of a high sacrifice rate at the FHCRC due to limited space and facilities. Cancer incidence times (ages) are thereby subject to substantially higher censoring rates in transplant than in control dogs.

Censored data rank tests are widely used to test equality of two or more sets of censored failure time data. Perhaps the two most frequently used tests are Gehan's (1965) general-

Key Words: Censored data rank tests; Logrank test; Generalized Wilcoxon tests; Proportional hazards model; Time-dependent covariates.

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ization of the Wilcoxon test (see also Breslow 1970) and Mantel's (1966) generalization of the Savage (1956) test, which is often referred to as the logrank test (Peto and Peto 1972). Like uncensored data rank statistics, these procedures assign scores to the ordered observations with all samples combined and form a test statistic by adding the scores corresponding to particular samples. The Gehan test applied to the beagle cancer incidence data gave a significance level of 0.76. In sharp contrast, the logrank test indicated an increased cancer incidence rate among the TBI treated dogs with a significance level of 0.01. The next section presents these statistics in a manner that explains how such remarkably different significance levels can arise.

2. *A Representation for Censored Data Rank Tests*

Let $t_1 < t_2 < \ldots < t_k$ denote the $k = 29$ ordered cancer incidence times for the combined $n = 281$ dogs. Let $z_1, z_2, \ldots, z_k$ be the corresponding group indicators with $z_i = 1$ or 0 according to whether or not a dog is a marrow transplant recipient. Also let $t_{il}, \ldots, t_{im(i)}$ denote the censoring times in $[t_i, t_{i+1})$, $i = 0, \ldots, k$ for all animals that either died naturally without cancer or by sacrifice or were alive at the time of data collection. Set $t_0 = 0$ and $t_{k+1} = \infty$. Let $z_{il}, \ldots, z_{im(i)}$ be the corresponding group indicators and set $s_i = z_{il} + \ldots + z_{im(i)}$, $i = 0, \ldots, k$. Censored data rank tests are then of the form

$$v = \sum_{i=1}^{k} (z_i c_i + s_i C_i), \tag{1}$$

where the constants c_i and C_i are scores corresponding to uncensored and censored failure times, respectively. The Gehan statistic assigns a score to a failure time, t, that equals the number of observations in the pooled sample known to be larger than t minus the number known to be smaller than t. This gives

$$c_i = n_i - i, \qquad C_i = -i, \qquad i = 1, \ldots, k, \tag{2}$$

where n_i is the number of study subjects at risk for cancer (without cancer and under active follow-up) just prior to time t_i (at time $t_i - 0$). If there is no censoring $n_i = n - i + 1$ and (1), with scores (2), is easily seen to reduce to the Wilcoxon statistic. The logrank test is given by scores

$$c_i = 1 - \sum_{j=1}^{i} n_j^{-1}, \qquad C_i = -\sum_{j=1}^{i} n_j^{-1}, \qquad i = 1, \ldots, k, \tag{3}$$

which reduces to one minus the expected order statistics from an exponential distribution if there is no censoring. Peto and Peto (1972) and Prentice (1978) give methods for generating classes of censored data rank statistics. Both methods give a generalization of the Wilcoxon statistic that differs from (2) and that have scores of the form

$$c_i = 2F_i - 1, \qquad C_i = F_i - 1, \tag{4}$$

where F_i is a survivor function estimator at time t_i. With Prentice's method this estimator is specified by

$$F_i = \prod_{j=1}^{i} [n_j/(n_j + 1)], \qquad i = 1, \ldots, k. \tag{5}$$

Once again it is easy to see that (1) with scores given by (4) and (5) reduces to the Wilcoxon statistic if there is no censoring. This statistic will be referred to as the Peto generalized Wilcoxon statistic.

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A revised presentation of these statistics, in terms of observed and conditionally expected numbers of cancers in the transplant group ($z = 1$), facilitates the comparison between statistics of the form (1). Set $w_i = n_i(C_{i-1} - C_i)$, $i = 1, \ldots, k$ with $C_0 = 0$. Substitution into (1) gives

$$v = \sum_{i=1}^{k} z_i(c_i - C_i) - \sum_{i=1}^{k} (z_i + s_i) \sum_{j=1}^{i} w_j n_j^{-1} = \sum_{i=1}^{k} z_i(c_i - C_i) - \sum_{j=1}^{k} w_j n_j^{-1} \sum_{i=j}^{k} (z_i + s_i)$$

$$= \sum_{i=1}^{k} z_i(c_i - C_i - w_i) + \sum_{j=1}^{k} w_j(z_j - n_{1j} n_j^{-1}), \quad (6)$$

where $n_{1j} = (z_j + s_j) + \ldots + (z_k + s_k)$ is the number of transplant dogs ($z = 1$) at risk at $t_j - 0$. Note that the second term in (6) is a weighted sum of the observed number, z_j, of cancers in the transplant group at t_j minus the conditionally expected number, $n_{1j} n_j^{-1}$, given the risk set sizes n_{1j} and n_j, under the null hypothesis. If the coefficients $c_i - C_i - w_i$, $i = 1, \ldots, k$ of the first term are identically zero then the statistic v has a particular convenient representation in terms of observed and 'expected' cancer incidence. This condition can alternatively be written

$$n_i C_{i-1} = c_i + (n_i - 1)C_i, \qquad i = 1, \ldots, k, \quad (7)$$

which can be interpreted in terms of preservation of the total sum of scores at each t_i. More specifically, the same sum of scores arises if follow-up, for each study subject at risk, were to terminate just prior to t_i or to terminate immediately after t_i. Expression (7) could be used to generate censored data test statistics, for example, by specifying c_i, $i = 1, \ldots, k$, perhaps on the basis of uncensored rank test scores expressed as a function of a survivor function estimator, and then using (7) to calculate corresponding C_i, $i = 1, \ldots, k$.

It is easy to verify that the three test statistics given by (2)–(5) all satisfy (7) so that they have a representation as

$$v = \sum_{i=1}^{k} w_i(z_i - n_{1i} n_i^{-1}) \quad (8)$$

where the respective weights $w_i = c_i - C_i$ are $w_i = n_i$, $w_i \equiv 1$, and $w_i = F_i$. The logrank statistic, as is well known, is simply the sum of observed minus conditionally expected cancers, Gehan's statistic weights the difference between observed and 'expected' numbers at t_i according to the size, n_i, of the risk set at t_i whereas the Peto Wilcoxon generalization, based on (4) and (5), weights these differences according to the survivor function estimator F_i.

With tied failure times (8) generalizes readily to

$$v = \sum_{i=1}^{k} w_i(d_{1i} - d_i n_{1i} n_i^{-1}) \quad (9)$$

where d_{1i} is the number of group 1 cancers at t_i and d_i is the total number of cancers at t_i. The hypergeometric distribution gives the null hypothesis conditional variance of each of the terms in the sum (9) and shows that the expectation of (9) is zero under the null hypothesis. An argument virtually identical to that used by Cox (1975), along with the fact that w_i is fixed, given the risk set sizes up to t_i, shows that the terms of (9) are uncorrelated. It follows that the sum of the w_i^2 times the corresponding hypergeometric variance gives a variance estimator for v. This estimator can be written

$$V = \sum_{i=1}^{k} w_i^2(n_{1i}/n_i)[(1 - n_{1i})/n_i][(n_i - d_i)/(n_i - 1)]d_1 \quad (10)$$

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where the final factor is defined to take value zero if $n_i = 1$. A standardized test statistic is then $vV^{-1/2}$. Note that with tied failure times (5) should be modified to

$$F_i = \prod_{j=1}^{i} [(n_j - d_j + 1)/(n_j + 1)]$$

in determining (9) and (10). Note also that a permutation approach to variance estimation is invalid with data sets in which the censoring rate differs among samples.

3. *Application to the Canine Transplant Data*

Table 1 lists the 29 cancer incidence ages arising from the sample of 281 dogs. Four of the 29 cancers occurred in transplant dogs. Table 1 also gives the total number of dogs and the number of transplant dogs at risk at each cancer incidence time. The heavy censoring in the transplant group is evident with only 10 transplant dogs followed to 2380 days and none followed beyond 3568 days. In comparison, 105 and 54 control dogs were at risk at these points in time. The fifth column of Table 1 gives the observed minus conditionally expected cancers at each cancer incidence time. The logrank test merely adds these differences. Gehan's statistic on the other hand weights these differences according to the corresponding risk set sizes. Because of the heavy censoring these weights decline rapidly (column 3), much more so than do the 'survivor function' weights (column 6), giving rise to the Peto generalized Wilcoxon statistic. The final three columns of Table 1 give the contributions at each failure time to the standardized statistic $vV^{-1/2}$ for each of the three statistics. Note that the logrank and Peto's Wilcoxon statistic agree closely corresponding to the fact that the weights F_i are between 1 and 0.860 at all cancer incidence times at which both transplant and control dogs are at risk. Note the large positive contributions to these two tests statistics at the four cancer incidence times among transplant dogs. These contributions are large since the control risk sets are very much larger than the transplant risk sets at each of these times. The standardized statistics have values 2.53 (logrank) and 2.42 (Peto's generalized Wilcoxon) which, assuming standard normality, are both significant at the 0.02 level. Gehan's generalized Wilcoxon statistic, on the other hand, assigns much smaller positive weight to the four cancers among transplant dogs since very substantial censoring has occurred by these time points. The result is a standardized statistic of value 0.31 which is not at all suggestive of a difference in cancer incidence rates.

In view of the relatively small number of time points that contribute to the calculation of the test statistic, the standard normal approximation to $vV^{-1/2}$ and the corresponding significance levels given may not be very accurate. Some computer simulation or more sophisticated approximation would be worthwhile. Such work, however, is unlikely to explain the discrepancy between the three statistics to any degree.

As noted above, permutation variance estimators are inappropriate with these data because of substantially different censoring patterns in the two groups. Such estimators are, in fact, highly inflated and give rise to standardized statistics for all three tests that are much reduced in absolute value.

4. *Generalized Wilcoxon Statistics*

The Peto generalized Wilcoxon statistic assigns weights that depend on the estimated percentile of the failure time (cancer incidence) distribution. Failures occurring early, when the estimated survivor function is large, receive large weights while those in the right tail of the failure time distribution receive small weights, as is the case with the uncensored data Wilcoxon statistic. Gehan's statistic, on the other hand, assigns weights that depend on the

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TABLE 1
Cancer Incidence Data and Test Statistic Calculation for Transplant and Control Dogs

Age (days) at cancer incidence (t)	Group: Transplant (1) Control (0) (z)	Dogs at risk (n)	Transplant dogs at risk (n_1)	Observed minus expected ($z-n_1 n^{-1}$)	Survivor estimator (5) (F)	Logrank contribution	Gehan's Wilcoxon contribution	Peto's Wilcoxon contribution
246	0	273	83	−.304	.996	−.29	−.37	−.30
322	0	272	83	−.305	.993	−.29	−.37	−.30
400	0	264	79	−.299	.989	−.29	−.35	−.29
2380	1	105	10	.905	.980	.87	.43	.88
2640	1	99	7	.929	.970	.89	.41	.89
2844	0	89	4	−.045	.959	−.04	−02	−.04
2858	0	88	4	−.046	.948	−.04	−.02	−.04
2995	0	86	4	−.047	.937	−.04	−.02	−.04
3088	1	76	4	.947	.925	.91	.32	.87
3100	1	75	3	.960	.913	.92	.32	.87
3168	0	74	2	−.027	.901	−.03	−.01	−.02
3267	0	70	1	−.014	.888	−.01	−.00	−.01
3364	0	69	1	−.015	.876	−.01	−.00	−.01
3454	0	57	1	−.018	.860	−.02	−.00	−.01
3568	0	54	0	0		0	0	0
3675	0	51	0	0		0	0	0
3738	0	50	0	0		0	0	0
3870	0	40	0	0		0	0	0
3961	0	39	0	0		0	0	0
4171	0	26	0	0		0	0	0
4321	0	24	0	0		0	0	0
4568	0	21	0	0		0	0	0
4674	0	20	0	0		0	0	0
4680	0	19	0	0		0	0	0
4711	0	17	0	0		0	0	0
5006	0	14	0	0		0	0	0
5011	0	13	0	0		0	0	0
5023	0	12	0	0		0	0	0
5603	0	2	0	0		0	0	0
Standardized Test Statistic $vV^{-1/2}$						2.53	0.31	2.42

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estimated percentile of the distribution for failures and censorings combined. Such dependence on censoring rates is inappropriate and can lead to anomalous situations. Specifically, this dependence permits the test statistic to be dominated by a small number of early failures regardless of the relationship between these failures and the underlying failure time distributions. A referee, Dr. Peter O'Brien, has offered an example which makes clear the inappropriateness of the Gehan statistic: Suppose n study subjects are followed until the first failure time, after which a random sample of m randomly selected subjects are followed until failure. For any specified m, the Gehan statistic will be completely dominated by the first failure time contribution as n becomes very large. In fact, as $n \to \infty$ the standardized statistic will approach $[(1-p)p^{-1}]^{1/2}$ or $-[p(1-p)^{-1}]^{1/2}$ depending on whether $z = 1$ or 0 at the first failure time, where p is the ratio of the group 1 risk set to the overall risk set at $t = 0$. For example, with equal sample sizes ($p = 1/2$) and for sufficiently large n, the standardized Gehan statistic will take values arbitrarily close to $+1$ or -1, and so will be nonsignificant, regardless of the relationship between the failure time distributions and regardless of the number, m, of study subjects that are followed until failure. With such a censoring scheme the Gehan statistic is biased ($p \neq 1/2$) and will have an asymptotic power that converges to zero regardless of differences between the failure rates being compared. The Peto statistic is not subject to such criticism and is the generalization of the Wilcoxon test that should be used with censored data.

5. *Further Insights into the Data*

The test statistics described above are all somewhat inappropriately applied, in that account is not taken of the waiting time to marrow transplantation. Dogs awaiting transplantation should properly be regarded as members of the control group until transplantation takes place. For example, the number of transplanted dogs at 246, 322 and 400 days, the first three cancer incidence times, is 33, 49 and 55, respectively, in contrast to the combined control group sizes of 240, 223 and 209. The fact that the three early cancers occurred in untransplanted dogs is no longer very surprising in view of these sample sizes.

The rank tests (9) and variance formulae (10) are readily modified to permit group membership to change with time. In fact, the same expressions apply but with n_{1i} representing the number of dogs that have received marrow transplants by time t_i, $i = 1, \ldots, k$. With logrank scores ($w_i \equiv 1$) this time-dependent statistic (see Mantel and Byar 1974) gives a difference between observed and expected cancers in the transplant group that totals to 3.08 (in comparison to 2.62 for the time-independent test) and an associated standardized statistic of 3.42. Comparison with standard normal tables gives a significance level of <.001, indicating that a sharper result has been obtained by taking account of waiting times to transplantation.

This time-dependent logrank test is, in fact, the score test for $\beta = 0$ from a proportional hazards model (Cox 1972) in which the cancer incidence rate at time t is specified as

$$\lambda_0(t) \exp \{z(t)\beta\} , \tag{11}$$

where $\lambda_0(t) \geq 0$ is an arbitrary function and $z(u)$, $u \geq 0$, is defined to have value zero until transplantation takes place and to have value 1 thereafter. The quantity e^{β} then represents the risk for cancer in a dog that has been transplanted relative to a dog of the same age that has not received a transplant. The parameter β can be estimated by applying asymptotic likelihood methods to the partial likelihood (Cox 1975),

$$L(\beta) = \prod_{i=1}^{k} \{\exp[z_i(t_i)\beta] \Big/ \sum_{\ell \in R(t_i)} \exp [z_\ell(t_i)\beta]\}$$

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where $R(t_i)$ consists of the set of animals at risk just prior to time t_i. A Newton-Raphson iteration gives $\hat{\beta} = 1.93$ so that the relative risk of cancer associated with marrow transplantation is estimated to be $\exp \hat{\beta} = 6.90$. An approximate 95% confidence interval for $\exp \hat{\beta}$ based on the asymptotic normal distribution for $\hat{\beta}$ is (1.97, 24.13). A significance test for $\beta = 0$ based on the asymptotic normal distribution of $\hat{\beta}$ gives a standard normal value of 3.02 in reasonable agreement with the corresponding score test (time-dependent logrank test). The adequacy of sample sizes for use of the asymptotic distribution of $\hat{\beta}$ is more questionable than is the case with the score test.

Additional time-dependent covariates allow the cancer risk associated with transplantation to depend in a smooth manner on the animal's age or other characteristics. For example, a specification $z(t) = \{z_1(t), z_2(t)\}$ where $z_1(t)$ is an indicator variable that takes value one following transplantation and value zero otherwise and $z_2(t) = z_1(t) \log t$ would allow the relative risk at age t, associated with transplantation, to be of the form $e^{\beta_1} t^{\beta_2}$. The parameter β_2 is, with these data, estimated by $\hat{\beta}_2 = 2.04$ giving rise, for example, to estimated relative risk estimates of 3.55, 5.59, 8.11 and 11.10 at 4, 5, 6 and 7 years of age, respectively. A test for $\beta_2 = 0$ has approximate significance level 0.16 from the asymptotic distribution of $\hat{\beta}_2$. In comparison the asymptotic significance level is less than 0.05 on the basis of the corresponding likelihood ratio test. With these sample sizes, however, such estimators and significance levels should only be taken to provide a mild indication that the relative risk of cancer associated with transplantation increases with age.

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